

Análisis de expresión diferencial - clase

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Librerías

```
# install.packages("BiocManager")

# install.packages("tidyverse")
# install.packages("ggplot2")
# BiocManager::install("DESeq2")
# BiocManager::install("clusterProfiler")

library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.1      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.2.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(DESeq2)
```

```
## Loading required package: S4Vectors
## Loading required package: stats4
## Loading required package: BiocGenerics
##
## Attaching package: 'BiocGenerics'
##
## The following objects are masked from 'package:lubridate':
##
##   intersect, setdiff, union
##
## The following objects are masked from 'package:dplyr':
##
##   combine, intersect, setdiff, union
##
```

```

## The following objects are masked from 'package:stats':
##
##   IQR, mad, sd, var, xtabs
##
## The following objects are masked from 'package:base':
##
##   anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##   colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##   get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
##   match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
##   Position, rank, rbind, Reduce, rownames, sapply, saveRDS, setdiff,
##   table, tapply, union, unique, unsplit, which.max, which.min
##
##
## Attaching package: 'S4Vectors'
##
## The following objects are masked from 'package:lubridate':
##
##   second, second<-
##
## The following objects are masked from 'package:dplyr':
##
##   first, rename
##
## The following object is masked from 'package:tidyr':
##
##   expand
##
## The following object is masked from 'package:utils':
##
##   findMatches
##
## The following objects are masked from 'package:base':
##
##   expand.grid, I, unname
##
## Loading required package: IRanges
##
## Attaching package: 'IRanges'
##
## The following object is masked from 'package:lubridate':
##
##   %within%
##
## The following objects are masked from 'package:dplyr':
##
##   collapse, desc, slice
##
## The following object is masked from 'package:purrr':
##
##   reduce
##
## Loading required package: GenomicRanges
## Loading required package: GenomeInfoDb

```

```

## Loading required package: SummarizedExperiment
## Loading required package: MatrixGenerics
## Loading required package: matrixStats
##
## Attaching package: 'matrixStats'
##
## The following object is masked from 'package:dplyr':
##
##     count
##
## Attaching package: 'MatrixGenerics'
##
## The following objects are masked from 'package:matrixStats':
##
##     colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
##     colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
##     colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
##     colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
##     colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
##     colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
##     colWeightedMeans, colWeightedMedians, colWeightedSds,
##     colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
##     rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
##     rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
##     rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
##     rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
##     rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
##     rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
##     rowWeightedSds, rowWeightedVars
##
## Loading required package: Biobase
## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages 'citation("pkgname)".
##
## Attaching package: 'Biobase'
##
## The following object is masked from 'package:MatrixGenerics':
##
##     rowMedians
##
## The following objects are masked from 'package:matrixStats':
##
##     anyMissing, rowMedians

```

```
library(clusterProfiler)
```

```

##
## clusterProfiler v4.14.6 Learn more at https://yulab-smu.top/contribution-knowledge-mining/
##

```

```
## Please cite:
##
## S Xu, E Hu, Y Cai, Z Xie, X Luo, L Zhan, W Tang, Q Wang, B Liu, R Wang,
## W Xie, T Wu, L Xie, G Yu. Using clusterProfiler to characterize
## multiomics data. Nature Protocols. 2024, 19(11):3292-3320
##
## Attaching package: 'clusterProfiler'
##
## The following object is masked from 'package:IRanges':
##
##     slice
##
## The following object is masked from 'package:S4Vectors':
##
##     rename
##
## The following object is masked from 'package:purrr':
##
##     simplify
##
## The following object is masked from 'package:stats':
##
##     filter
```

```
library(msigdb)
```

Cargar datos

```
rna <- read.table(file="../input/coad_rna.tsv", sep="\t", header=TRUE, check.names = FALSE, row.names =
metadata <- read.table(file="../input/coad_metadata.tsv", sep="\t", header = TRUE)

head(metadata)
```

```
##      PATIENT_ID      SAMPLE_ID SUBTYPE AGE  SEX  OS_STATUS
## 1 TCGA-A6-5667 TCGA-A6-5667-01    CIN  40 Female  0:LIVING
## 2 TCGA-AY-4070 TCGA-AY-4070-01    CIN  50 Female  1:DECEASED
## 3 TCGA-CM-6168 TCGA-CM-6168-01     GS  84 Female  0:LIVING
## 4 TCGA-CA-6716 TCGA-CA-6716-01    CIN  65   Male  0:LIVING
## 5 TCGA-F4-6569 TCGA-F4-6569-01    CIN  60   Male  0:LIVING
## 6 TCGA-A6-A56B TCGA-A6-A56B-01    CIN  57   Male  1:DECEASED
##  COUNTRY_OF_RESIDENCE  SITE_OF_BIOPSY purity ploidy wholeGenomeDuplication
## 1                <NA>      Sigmoid colon  0.69  3.20                true
## 2                <NA>       Cecum      0.31  4.39                true
## 3      United States  Ascending colon  0.35  2.17                false
## 4      Vietnam      Ascending colon  0.73  3.74                true
## 5      Russia      Transverse colon  0.40  2.16                false
## 6      United States  Sigmoid colon  0.72  3.21                true
##      CMS metastasis
## 1      CMS2      false
## 2 no_label      false
## 3      CMS4      false
```

```
## 4      CMS2      false
## 5      CMS4      false
## 6 no_label      false
```

```
dim(rna)
```

```
## [1] 60660   300
```

Extraer el dato a comparar

```
# El modelo DESeq2 necesita una colData que tenga en las filas las muestras, y la columna con el valor
```

```
coldata <- metadata %>% select(PATIENT_ID, wholeGenomeDuplication) %>%
  column_to_rownames(var = "PATIENT_ID") %>%
  mutate(wholeGenomeDuplication = as.factor(wholeGenomeDuplication))
```

```
table(coldata)
```

```
## wholeGenomeDuplication
## false  true
##   178   122
```

```
# eliminar genes sin expresion
```

```
table(rowSums(rna) > 0) # TRUE genes con expresion, FALSE genes donde en todas las muestras son 0
```

```
##
## FALSE  TRUE
## 12252 48408
```

```
rna <- rna[rowSums(rna) > 0, ]
dim(rna)
```

```
## [1] 48408   300
```

Correr análisis de expresión

```
dds <- DESeqDataSetFromMatrix(countData = rna,
                              colData = coldata,
                              design = ~wholeGenomeDuplication)
```

```
dds <- DESeq(dds)
```

```
## estimating size factors
```

```
## estimating dispersions
```

```
## gene-wise dispersion estimates

## mean-dispersion relationship

## final dispersion estimates

## fitting model and testing

## -- replacing outliers and refitting for 1599 genes
## -- DESeq argument 'minReplicatesForReplace' = 7
## -- original counts are preserved in counts(dds)

## estimating dispersions

## fitting model and testing
```

```
resultsNames(dds)
```

```
## [1] "Intercept"
## [2] "wholeGenomeDuplication_true_vs_false"
```

```
res <- results(dds, name = "wholeGenomeDuplication_true_vs_false") %>% as.data.frame()
summary(res)
```

```
##      baseMean      log2FoldChange      lfcSE      stat
## Min.      : 0.00      Min.      :-4.88285      Min.      :0.0000      Min.      : -8.76153
## 1st Qu.: 0.02      1st Qu.: -0.11598      1st Qu.: 0.1368      1st Qu.: -0.27856
## Median : 0.21      Median : -0.02934      Median : 0.4921      Median : -0.01438
## Mean      : 20.69      Mean      : -0.03760      Mean      : 1.2213      Mean      : -0.05983
## 3rd Qu.: 4.30      3rd Qu.: 0.04479      3rd Qu.: 2.9468      3rd Qu.: 0.09226
## Max.     :33423.21      Max.      : 5.25742      Max.      : 2.9468      Max.      :10.86909
##
##      pvalue      padj
## Min.      :0.0000      Min.      :0.000
## 1st Qu.:0.2494      1st Qu.:0.132
## Median :0.8512      Median :0.454
## Mean      :0.6435      Mean      :0.458
## 3rd Qu.:0.9841      3rd Qu.:0.765
## Max.      :1.0000      Max.      :1.000
##
##      NA's      :25371
```

```
# log2FC > 0 - enriquecido en true, < 0 enriquecido en false
# Un log2FC de 1 indica un cambio de dos veces. Es decir, si es 1 entonces en esa condicion esta expres
true_enriched <- res %>% filter(log2FoldChange > 1, padj < 0.1)
false_enriched <- res %>% filter(log2FoldChange < -1, padj < 0.1)

cat("Hay", nrow(true_enriched), "genes sobreexpresados en duplicacion de genoma completo, versus", nrow
```

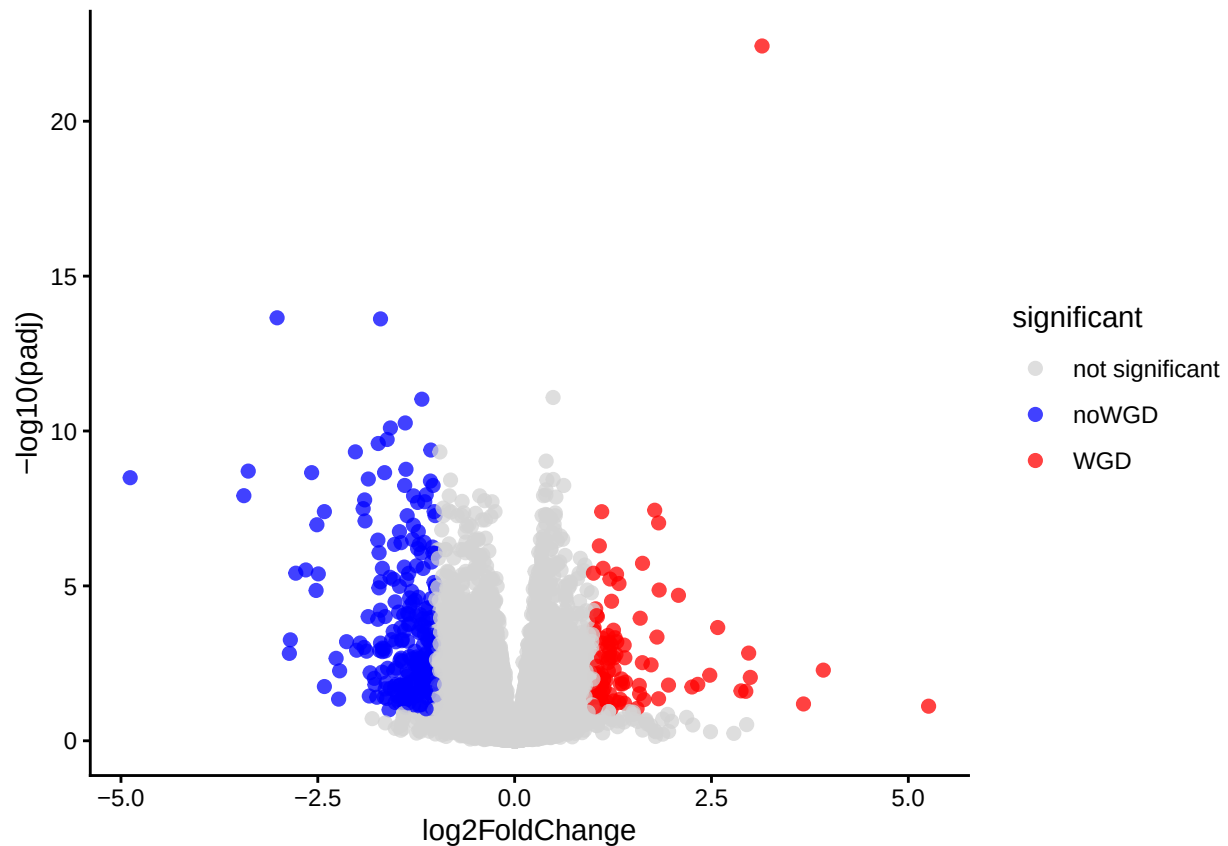
```
## Hay 101 genes sobreexpresados en duplicacion de genoma completo, versus 246 subexpresados.
```

Volcanoplot

```
# Definir colores
res <- res %>%
  mutate(gene = rownames(res),
         significant = ifelse(gene %in% rownames(true_enriched), "WGD",
                             ifelse(gene %in% rownames(false_enriched),
                                    "noWGD",
                                    "not significant"))))

# generar volcano
ggplot(res, aes(x = log2FoldChange, y = -log10(padj),
               color = significant)) +
  geom_point(alpha = 0.75, size = 2) +
  scale_color_manual(values = c("WGD" = "red",
                              "noWGD" = "blue",
                              "not significant" = "lightgray")) +
  theme_classic()
```

```
## Warning: Removed 25371 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```

# Definir una base de datos. Por ejemplo = GO procesos biológicos
go_bp <- msigdb(species = "Homo sapiens",
               collection = "C5",
               subcollection = "GO:BP")

# El universo de genes a estudiar debe ser los genes utilizados en la prueba
universo <- res$gene

# Se establece una tabla de asociacion proceso-gen
term2gene <- unique(as.data.frame(go_bp[, c("gs_name", "ensembl_gene")]))

```

```

# lista de genes con grupo
genes <- list("WGD" = res$gene[res$significant == "WGD"],
             "noWGD" = res$gene[res$significant == "noWGD"])

enrich <- compareCluster(geneClusters = genes,
                        fun = "enricher",
                        TERM2GENE = term2gene,
                        universe = universo,
                        pAdjustMethod = "BH",
                        pvalueCutoff = 0.1,
                        minGSSize = 3)

```

```

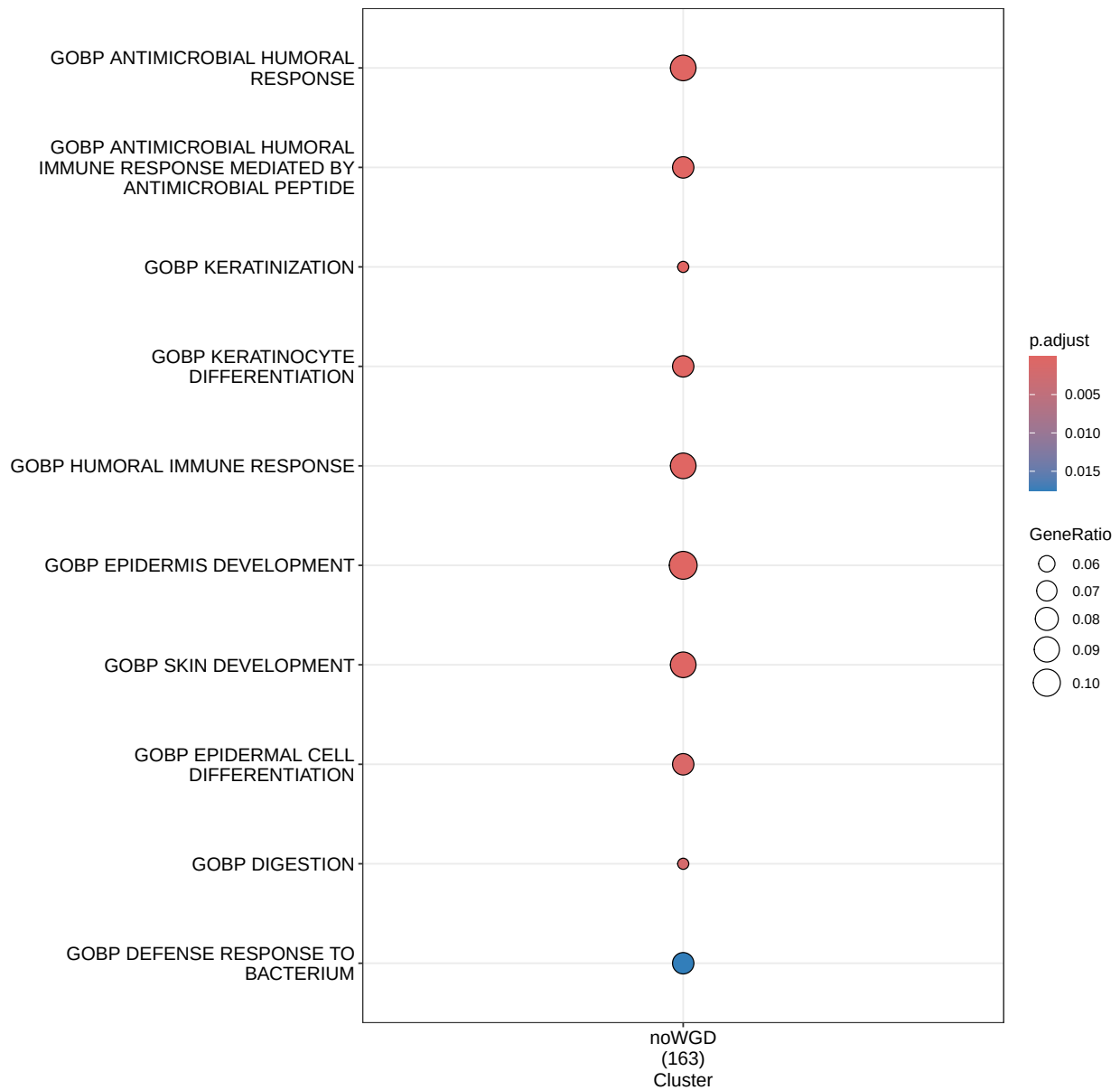
dotplot(enrich, showCategory = 10,
        color = "p.adjust")

```

```

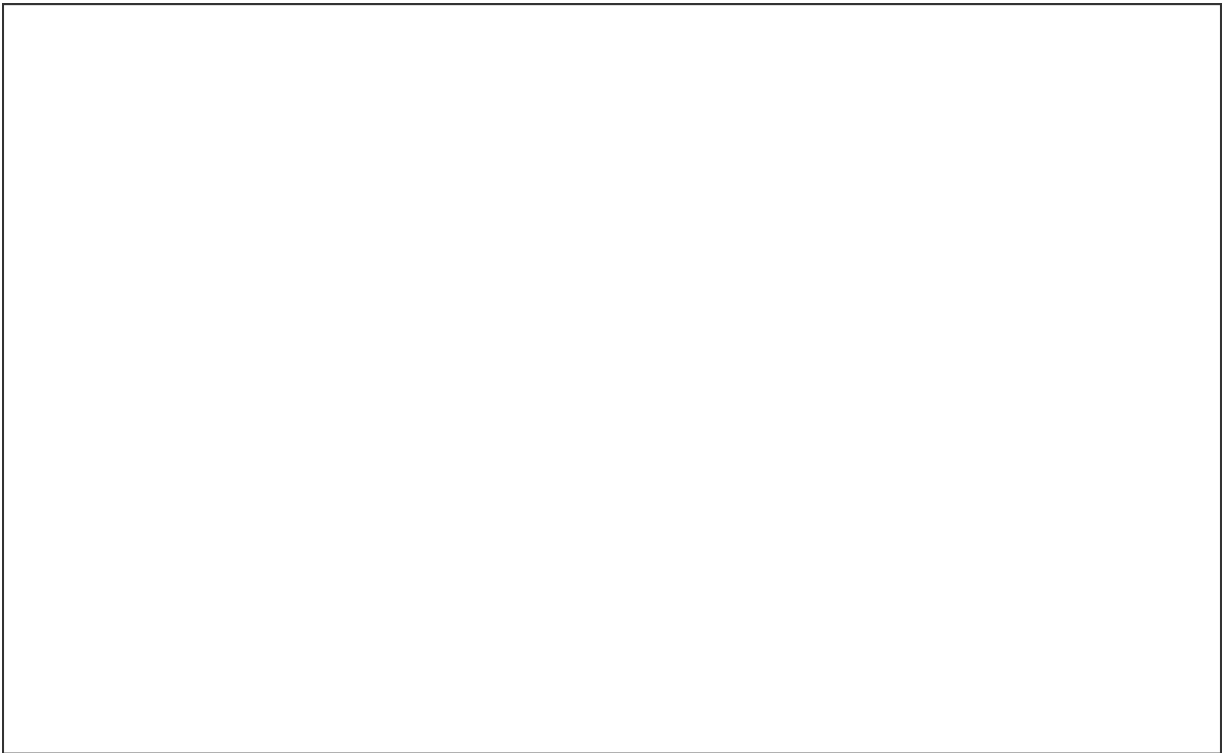
## Warning: 'aes_string()' was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with 'aes()'.
## i See also 'vignette("ggplot2-in-packages")' for more information.
## i The deprecated feature was likely used in the enrichplot package.
##   Please report the issue at
##   <https://github.com/GuangchuangYu/enrichplot/issues>.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```



Alternativa: un enriquecimiento por categoria

```
wgd_enrich <- enricher(gene = genes$WGD, TERM2GENE = term2gene,
                        universe = universo, pAdjustMethod = "BH",
                        pvalueCutoff = 0.1, minGSSize = 3)
dotplot(wgd_enrich, showCategory = 10,
        color = "p.adjust")
```



GeneRatio

```
nowgd_enrich <- enricher(gene = genes$noWGD, TERM2GENE = term2gene,  
                          universe = universo, pAdjustMethod = "BH",  
                          pvalueCutoff = 0.1, minGSSize = 3)  
dotplot(nowgd_enrich, showCategory = 10,  
        color = "p.adjust")
```

